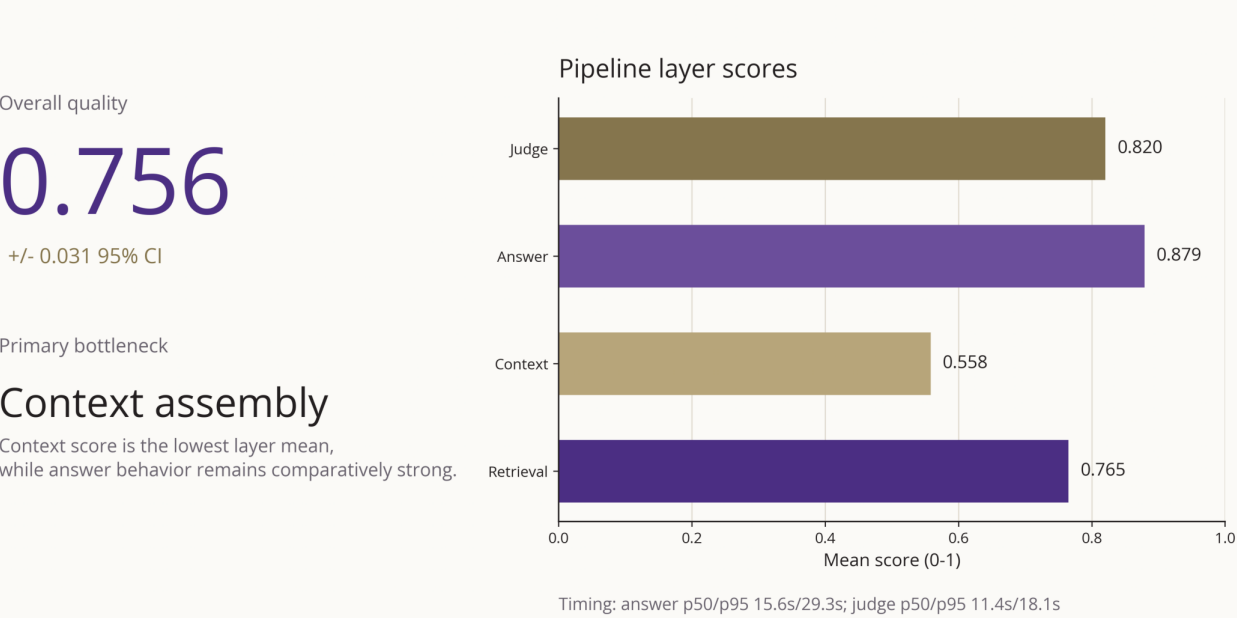


01 pipeline overview

Hard RAG benchmark: current system state

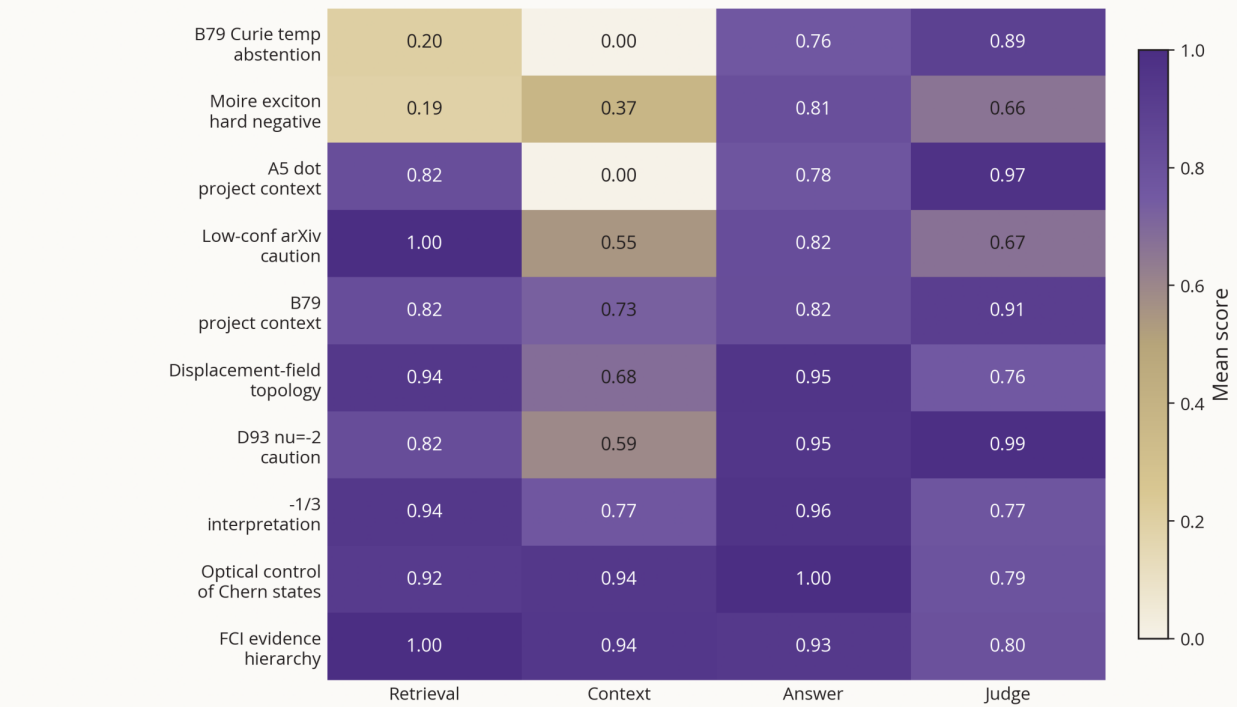
100 runs across 10 hard scientific tasks; error bars show repeat variability where applicable.



03 pipeline layer heatmap

Where performance is won or lost

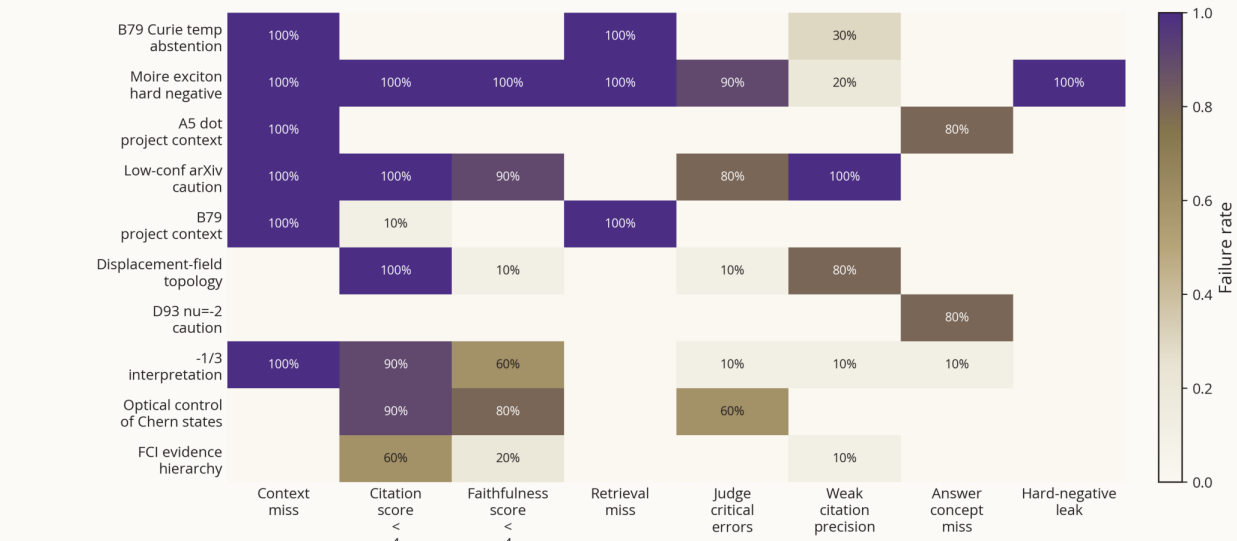
Rows are tasks sorted by overall quality; columns split the RAG pipeline.



05 failure by task heatmap

Failure rates by task

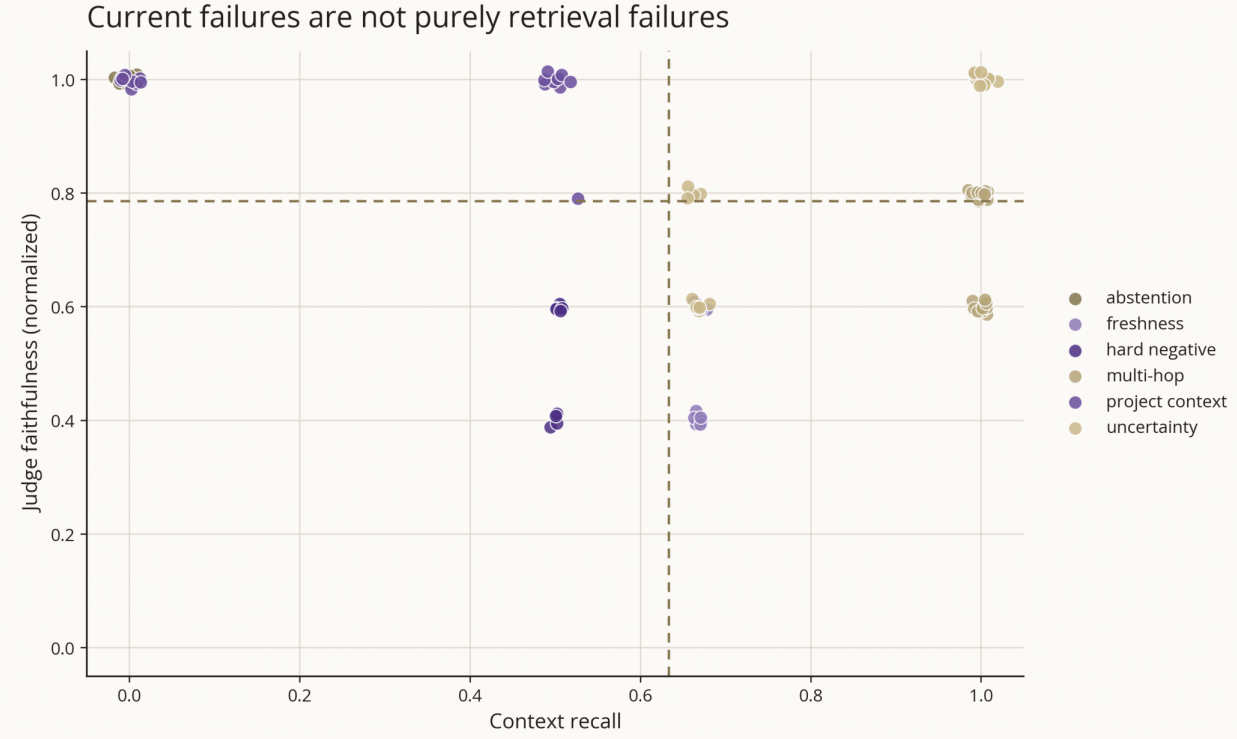
Darker cells identify systematic failure modes, not one-off noise.



07 context recall vs faithfulness

Context recall vs. answer faithfulness

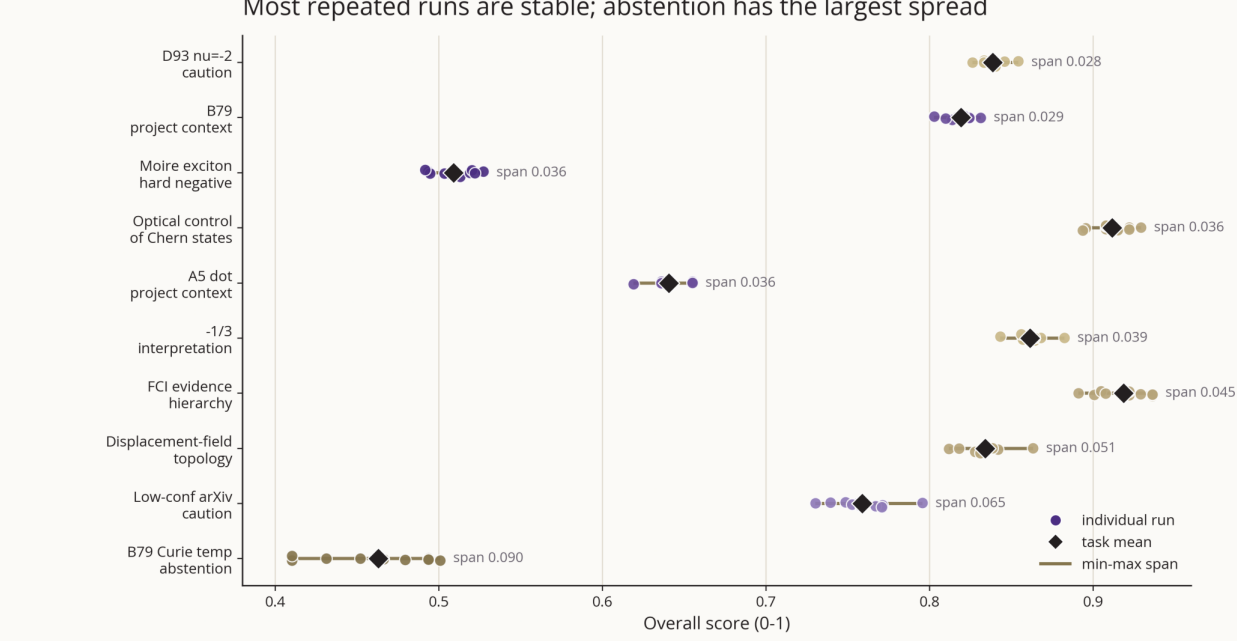
Each dot is one run. High recall without faithfulness points to answer/citation issues; low recall bounds answer quality.



09 repeatability run scores

Repeatability across runs

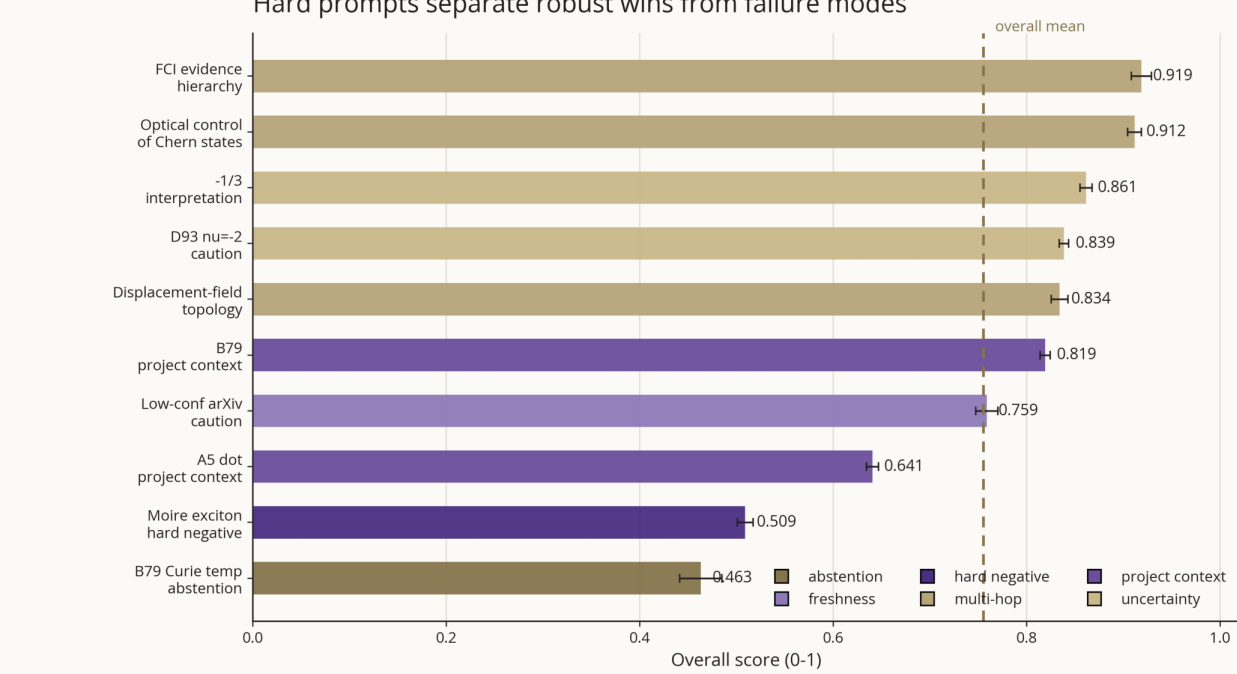
Each dot is one Codex answer+judge run. Horizontal bars show min-to-max; diamonds show the task mean.



02 task quality scorecard

Task-level quality

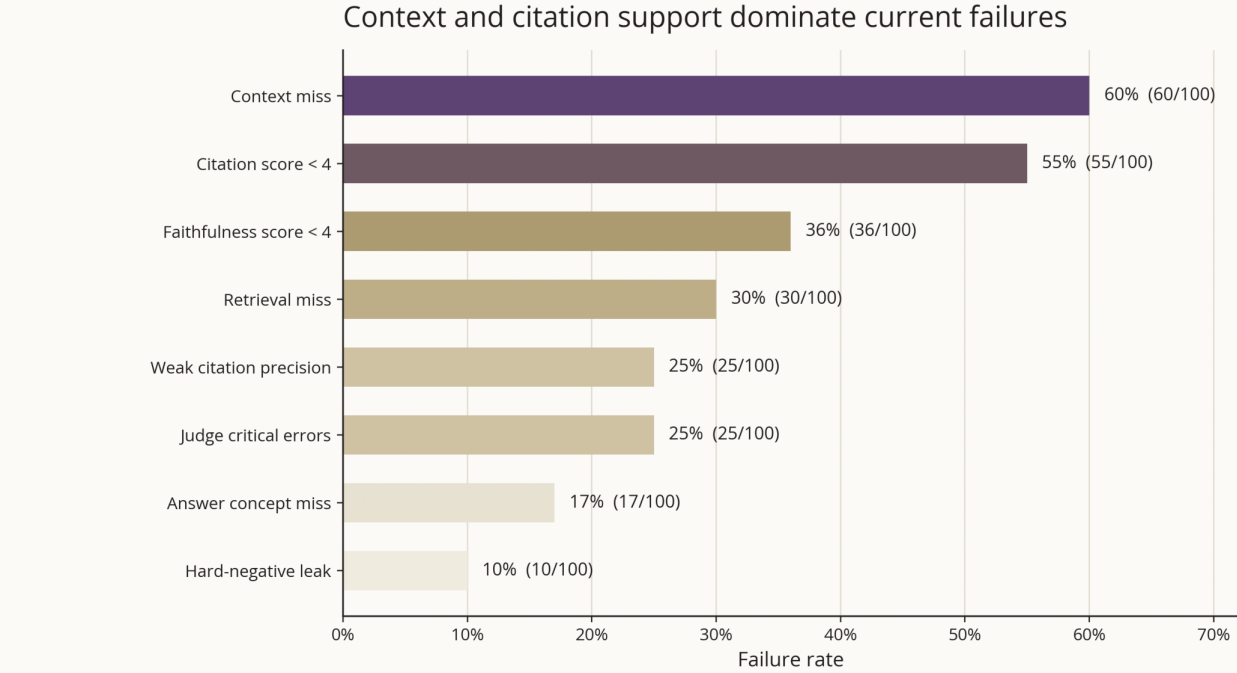
Mean overall score over 10 repeats per task; whiskers are 95% confidence intervals.



04 failure taxonomy

Failure taxonomy

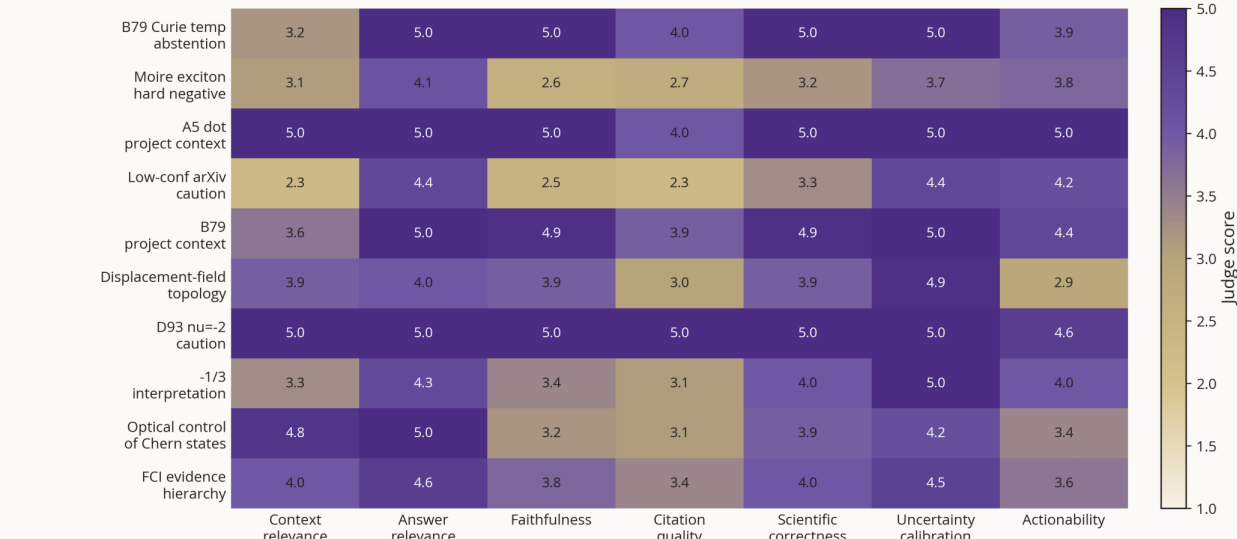
Run-level failure rates across the 100 hard RAG trials.



06 judge metric heatmap

LLM judge score profile

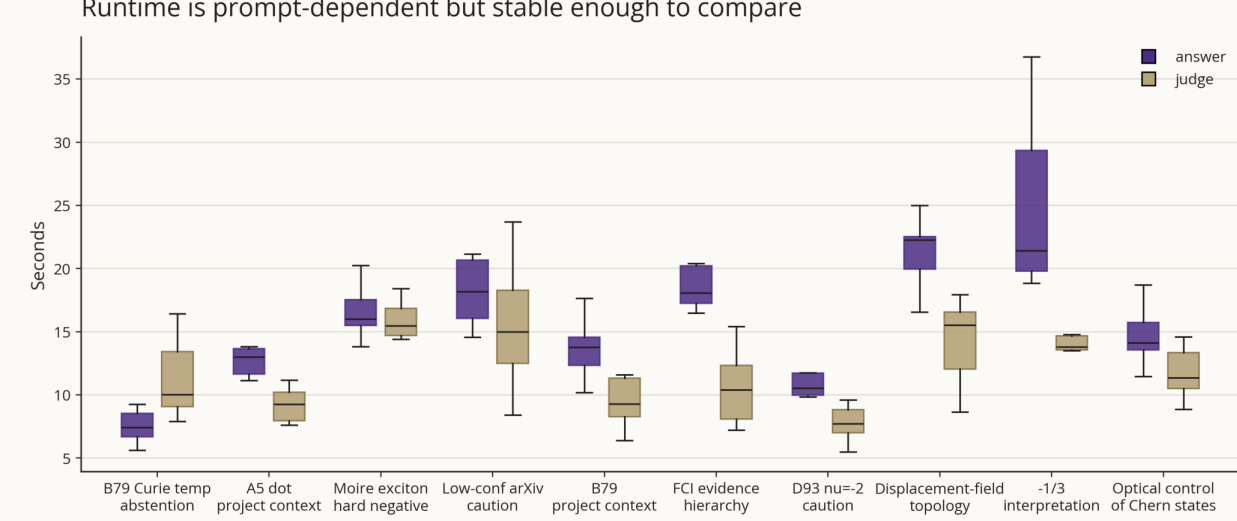
Scores are 1-5; citation quality and faithfulness are the weak judge dimensions.



08 latency by task

Codex timing by task

Answer and judge latency distributions over 10 repeats per hard prompt.



10 repeatability variation

How much do repeated scores move?

Tasks ranked by max-min score span across 10 independent runs.

